FINITE PRESENTATION RELATIVE TO A BASE IS CHECKABLE AFTER PUSHFORWARD ALONG A FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16472-16479 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16481-16485, the complete author proof (3 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Companion to wave 14's finite-relative-presentation dictionary cards (L16126/L16130 extend family): pushforward-side test here.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma more-algebra-lemma-finite-extension.

Source reference. Original lemma 05H0: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16485–16485. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite morphism of schemes locally of finite type over a base scheme S. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Then $\mathcal{F}$ is of finite presentation relative to S if and only if $\pi_*\mathcal{F}$ is of finite presentation relative to S.*

Proof. Translation of the result of More on Algebra, Lemma 05H0 into the language of schemes. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.